

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2026

Programme: BE

Full Marks : 100

Course: Digital Signal Analysis and Processing (New)

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define unit step signal. List out the Element of Digital Signal Processing(DSP) along with appropriate Diagram and Explain. 7

OR

What are the classification of Signal with examples in details?

- b) What is causal and non-causal system? Determine the output response of LTI system whose impulse response $h[n] = \{2,4,6,1\}$ and input $x[n] = \{1, 0,1, 1,1,0\}$. Verify the answer. 8

2. a) A discrete time signal is given by 8

$$h(n) = (-0.6)^n u(n) - 3(0.25)^n u(n)$$

Find:

- The Z-transform
- Region of Convergence (ROC)
- State whether the corresponding system is stable or not.

OR

Prove the Convolution property of the Z-transform. Explain why this property is useful in analyzing Linear Time-Invariant (LTI) systems.

- b) An LTI system is governed by 7

$$y(n) - 1.4y(n-1) + 0.49y(n-2) = x(n) + 0.5x(n-1)$$

- Obtain the transfer function
- Plot the poles and zeroes on the z-plane
- Determine whether the system is stable or not with justification.

3. a) Realize the following system by its direct form I and form II structure. 7

$$y[n] = x[n] - 2x[n-1] + 0.5x[n-2] + 0.8y[n-2]$$
- b) Determine the lattice coefficients corresponding to the FIR filter with system function and also draw the lattice structure. 8

$$H(z) = \frac{1}{1 - 0.9z^{-1} + 0.64z^{-2} - 0.576z^{-3}}$$
4. a) Let the given Equation $H_a(s) = \frac{s+0.2}{(s+0.2)^2+9}$ at $\omega = \frac{\pi}{2}$; then Find $H(z)$ using both IIM and BLT method. 7
- b) What are the steps for Butterworth filter design with correct order forms? Compute the Order of Filter having following specification: 8
 Passband frequency: 450 Hz
 Stopband frequency: 1000 Hz
 Passband attenuation: 3 dB
 Stopband attenuation: 10 dB
 Sampling frequency: 4000 Hz
5. a) Explain the Matched Z-Transformation method for digital IIR filter. Compare it with Impulse Invariant Method. 7
- b) Design a Chebyshev Type-I digital low-pass filter using the Bilinear Transformation method for the following specifications: 8
 Passband edge frequency = 1 kHz
 Stopband edge frequency = 1.8 kHz
 Passband Ripple = 1 dB
 Stopband Attenuation = 45 dB
 Sampling Frequency = 10 kHz
6. a) Find the Discrete Fourier Transform of the given discrete signal using formula method $x(n) = \{2, 0, 8, 4\}$ and verify the result with matrix method. 7
- b) Find 8-point DFT of the sequence. $\{0, 4, 1, 1, 2, 0, 2, 6\}$ using radix 2 DITFFT. 8
7. Write short notes on: (Any two) 2×5
 a) Sampling and its significance
 b) Properties of z-transform
 c) Linear phase of LTI system